The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF).

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Distributed Multi-Directory Agent Initializations

Each independent neuro-symbolic agent $a_k \in \mathcal{A}$ maintains its unique background logic constraints inside a dedicated disk directory path:

$$a_k \equiv \langle \mathcal{V}_k, \mathcal{K}_k, \mathcal{F}_k \rangle \tag{2}$$

4 Intensional Model Interrogation via Classification Type-Label Stripping

We formalize the mechanism for interrogating a frozen decision forest model by distinguishing between an exemplar's extensional type label and its structural attributes.

4.1 The Label Stripping Transformation Operator

Let $C_1 = \langle \text{namespace}, E_1, E_2, E_3, \pi_1, \pi_2, \pi_3, \mathcal{L}_{\text{type}}, \pm \rangle$ represent a complex hierarchical exemplar pattern, where $\mathcal{L}_{\text{type}}$ defines the explicit extensional classification type name (e.g., `compute_event`). To interrogate the forest without token name collisions or tautological loops, the system defines an intensional stripping operator $\mathcal{S}_{\text{strip}}$:

$$\mathcal{S}_{\text{strip}}(C_1) \implies \boldsymbol{v}_{\text{query}} = \langle E_1, E_2, E_3, \pi_1, \pi_2, \pi_3 \rangle \tag{3}$$

The resulting input vector $\boldsymbol{v}_{\text{query}}$ retains only the structural features, while the literal type label $\mathcal{L}_{\text{type}}$ is entirely withheld from the decision tree.

4.2 Non-Terminal Feature Vertex Traversal

The non-terminal split vertices within the decision forest $\mathcal{T}_k$ do not test or evaluate the type name label. Instead, the blind query vector $\boldsymbol{v}_{\text{query}}$ is passed down the geometric branch paths by evaluating purely structural thresholds:

$$f(\boldsymbol{v}_{\text{query}}) \implies \mathcal{E}_{\text{leaf}} \tag{4}$$

If the un-labeled feature pattern maps cleanly to the expected terminal leaf node context, the underlying structural isomorphism is verified across the agent poset layers. If a discrepancy or boundary exception is triggered at the leaf node ($\mathcal{E}_{\text{leaf}} \neq \mathcal{L}_{\text{type}}$), the new categorization label is fed back into the training exemplars for real-time model revision and ontology realignment.